

BINF 4002

Lecture 2: Linear Algebra

Updates

1. Approximate Syllabus is on Courseworks
2. V1 of answers for something including the 11 questions from the initial assignment also on Courseworks

Today's Class: Vector /
Vector Space / Linear
Operator / Matrix?

Outline

1. Why do we care (preview)?
2. Linearity
3. Vector Spaces
4. Linear Operators
5. Bases and Coordinatization
6. Why do we care?

Matrix Multiplication

$$\begin{array}{ccc} \text{matrix} & & \text{vector} \\ \begin{bmatrix} 1 & 4 & 3 \\ 3 & 1 & 2 \\ 1 & 0 & 6 \end{bmatrix} & \times & \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 8 \end{bmatrix} \end{array}$$

The components of $\mathbf{x}$ have the scalars in the linear combination of the columns of $\mathbf{A}$

$$x_1 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 8 \end{bmatrix}$$

Additional Things to Look Up

1. Eigen values and vectors
2. Singular values and vectors
3. Matrix vector derivative identities (e.g., what is the derivative of Trace)?
4. Matrix exponentiation